Lesson Plan · Thursday, November 5

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 1: Solving Linear Equations (1 day) · aimed at the Wed 11/18 test · essentials only (✪)

Objectives (students will…)

- Solve linear equations: distribute (incl. a **negative**), decimals, fractions, variables on both sides.
- Name the property for each step; check by substituting.
- Classify one / no / infinite solutions and **explain why** in a sentence.

TEKS

- **A.5A** (readiness) solve linear equations in one variable · Process: A.1D, A.1F, A.1G
- **Test link:** End of Topic Q2 (no solution, MC), Q4 (decimals), Q5 (explain one/none/infinite), Q11 (equal areas) = 4 of 11 items

Key terms

- solution · no solution · infinite solutions · properties of equality

Materials

- Slides: m3t1-lesson1-solving-linear-equations (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L1_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Thu-11-05.pdf + key (online: m3t1-lesson1-bellwork-day1)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Thu-11-05.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson1-Solving-Linear-Equations.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- SE pp. 345–352 · properties anchor chart (SE pp. 347–348) · whiteboards · algebra tiles

Bell work (on the board)

- Marcus has $35 and saves $15 per week. Which equation gives the amount he has saved, y, after x weeks? D. $y = 15x + 35$
- What value of x makes $3(x - 2) = 12$ true? F. 6
- Is $x = 2$ a solution of $5x - 1 = 9$? B. Yes, because $9 = 9$
- Which expression is equivalent to $4(x + 2) - x$? H. $3x + 8$

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	4 test-style items (A–D / F–J). Go over Q2 (distribute first) and Q3 (substitution checks a solution). Tell students: **4 of the 11 questions on the 11/18 test come from today.**
5–15	✪ Getting Started: Equation Creation · SE p. 346	Definition of *solution*. Pairs build from $x = 2$: $x + 3 = 5 \rightarrow 4x + 12 = 20 \rightarrow 2x + 12 = -2x + 20 \rightarrow x + 6 = 10 - x$; swap and check $x = 2$. **Look for:** multiplying only the x-term instead of *every* term (the same error that sinks test Q2 and Q5).
15–32	✪ Activity 1.1: Using Properties to Justify Solutions · SE pp. 347–348	Explicit teach: properties of equality table. Q1 Harper and Sofia both give $x = 2$ ($8 = 8$, $4 = 4$). Q2 pairs, name each property: (b) $0.6(2x + 1) = 4x + 6.2$ $x = -2$ **(decimal, like test Q4)**; (c) $\frac{1}{2}x - 6 = 2 + (2x + 1)$ $x = -6$. Q3 Daniel vs. Chloe, $4(3x + 2) = 8x + 4$ $x = -1$: distribute first or divide by 4 first, both correct.
32–42	✪ Activity 1.2: Solutions to Linear Equations · SE pp. 349–350	$2 = 2$ stays true (infinite); $2 = 3$ stays false (no solution). Q4 table: $2x = 3x$ one solution, $x = 0$. Board rule, **simplify each side first**: same x-term + same number → infinite · same x-term + different number → none · different x-terms → one.
42–50	Test moves (publisher slides 35–36 + Talk the Talk 45–48)	**Negative outside ( ):** $6 - 4(m + 2) = 3 - 2(2m + 1) \rightarrow -4m - 2 = -4m + 1$ no solution (like Q2). **Explain your reasoning** (Q5): $\frac{1}{4}(8x + 12) = 2(x + 2) - 1 \rightarrow 2x + 3 = 2x + 3$ infinite. Sentence frame on the board: "The equation has ___ solution(s) because both sides simplify to ___." **Equal areas** (Q11, Talk the Talk SE p. 352): $6(5 + 2x) = 8(x + 5)$ $x = 2.5$. Write the area formula first; triangle = ½·b·h.
50–58	Exit ticket (independent, like test Q2, Q4, Q5, Q11)	1. Solve: $7 - 3(p + 2) = 10 - (2p + 5)$ -4 2. Solve: $0.5(4x - 6) = -1.5x + 4$ $x = 2$ 3. One, none, or infinite? $\frac{1}{2}(8x + 6) = 2(2x + 1) + 2$ No solution 4. Triangle: base 6, height $(x + 5)$. Rectangle: 4 by $(x - 1)$. Equal areas. Find x. $x = 19$ IEP/504: read aloud on request, properties/notes card, extra time.
58–60	Wrap up	Essential Question (SE p. 345). Homework: SE pp. 355–356 (Practice 1a–e, 2a–f). Absent or stuck: Lesson 1 video. Next class Mon 11/9: Lesson 2, Literal Equations.

Check for understanding

- Whiteboards: "$-4(m + 2) = ?$" $-4m - 8$ (sign check before the test moves)
- Thumbs: "$x + 5 = x + 5$ has no solution." Down: infinite

After class

- Exit Q1–2 missed → Skills Practice Set A; Q3 → Set B (list names: first small group on 11/17); Q4 → Set C + video.

Two weeks to the test (Wed 11/18). Lesson 1 skills come back as Q2, Q4, Q5 and Q11 of the 11/17 review. Skipped: Activity 1.3 Tic-Tac-Bingo (not starred; used 11/19). TE plans 2 days, district gives 1.